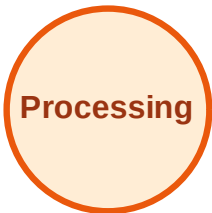


BFS Traversal

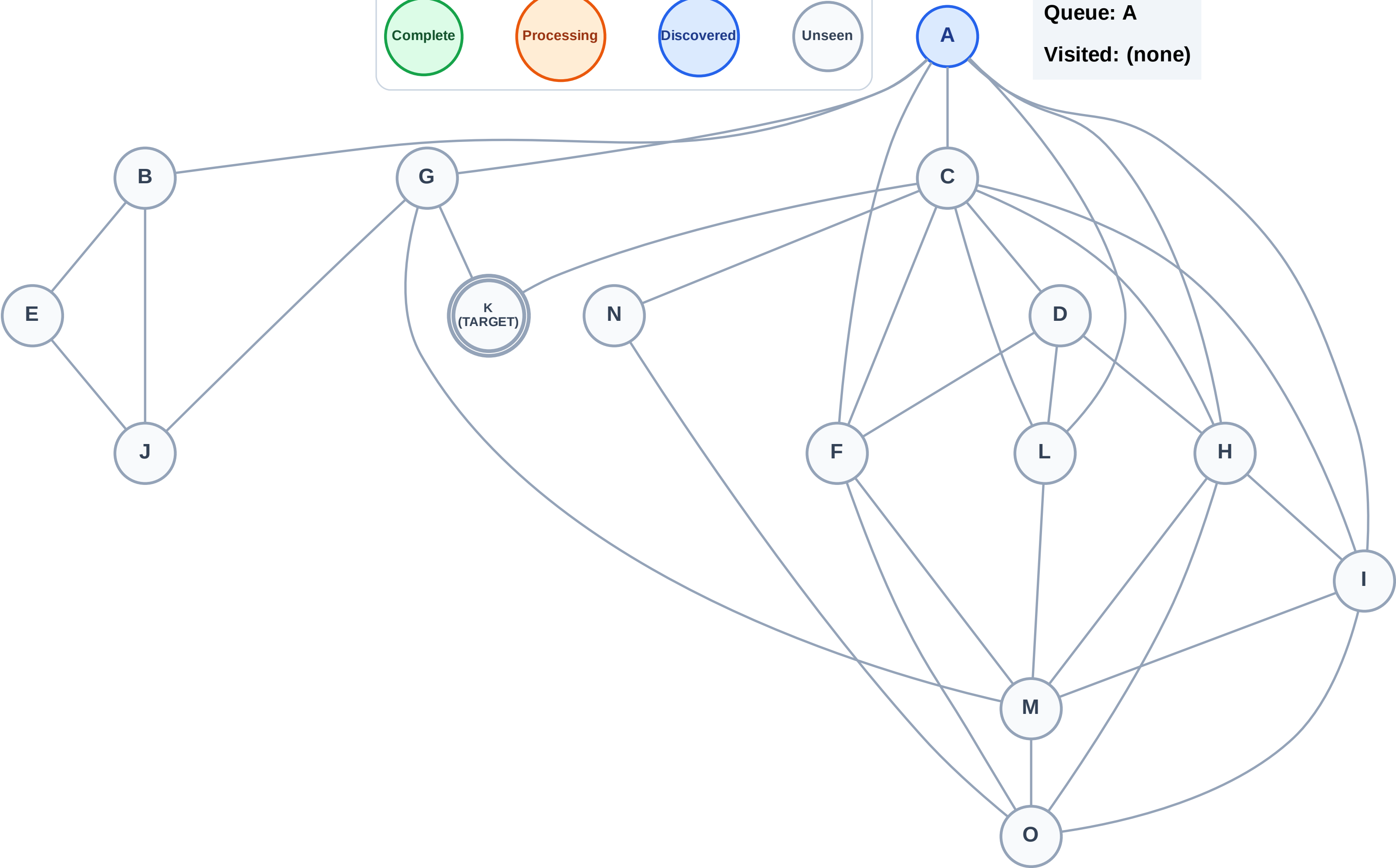
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Queue: A

Visited: (none)



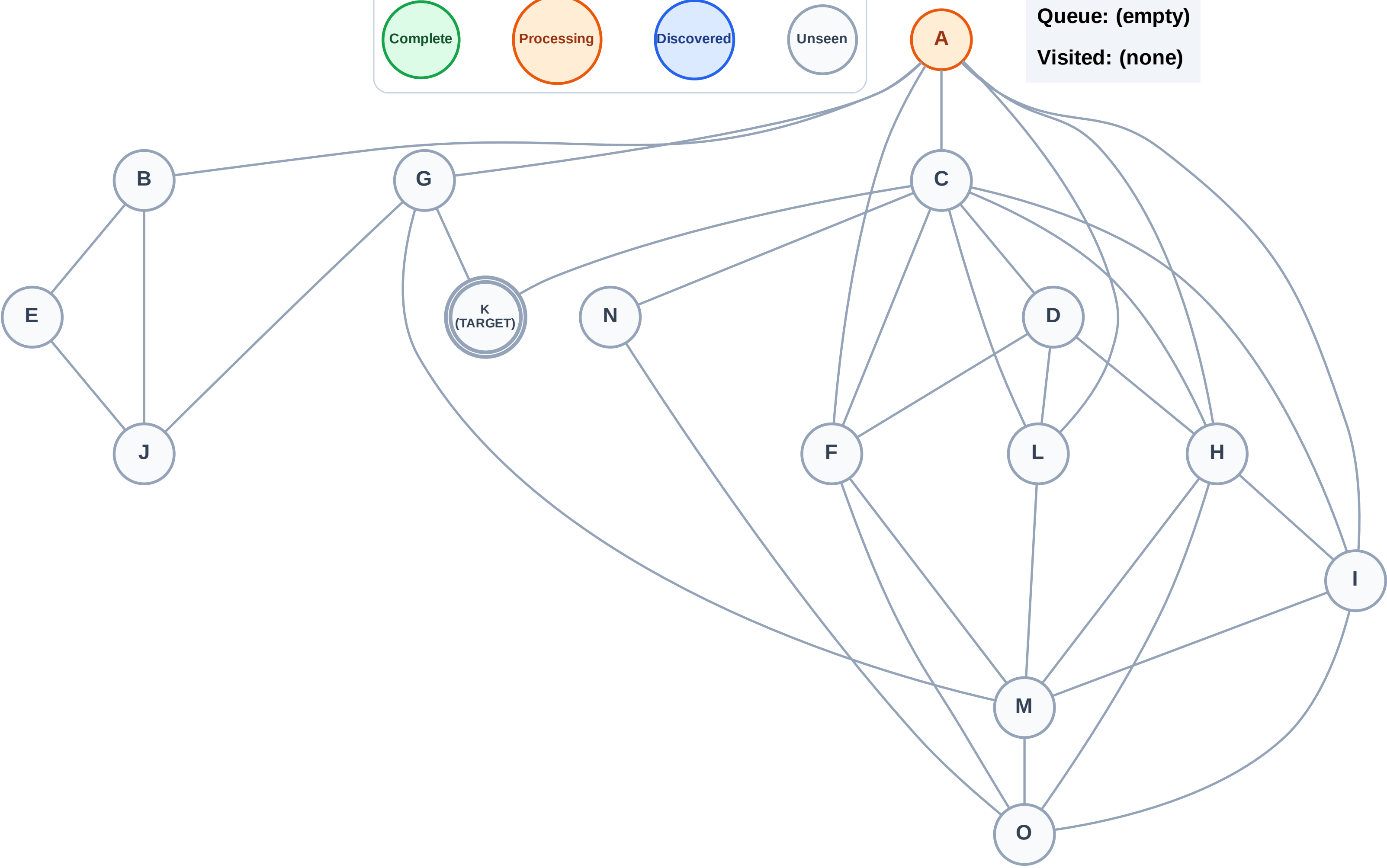
BFS Traversal

Step 2: Process node A

Legend



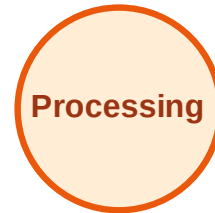
Queue: (empty)
Visited: (none)



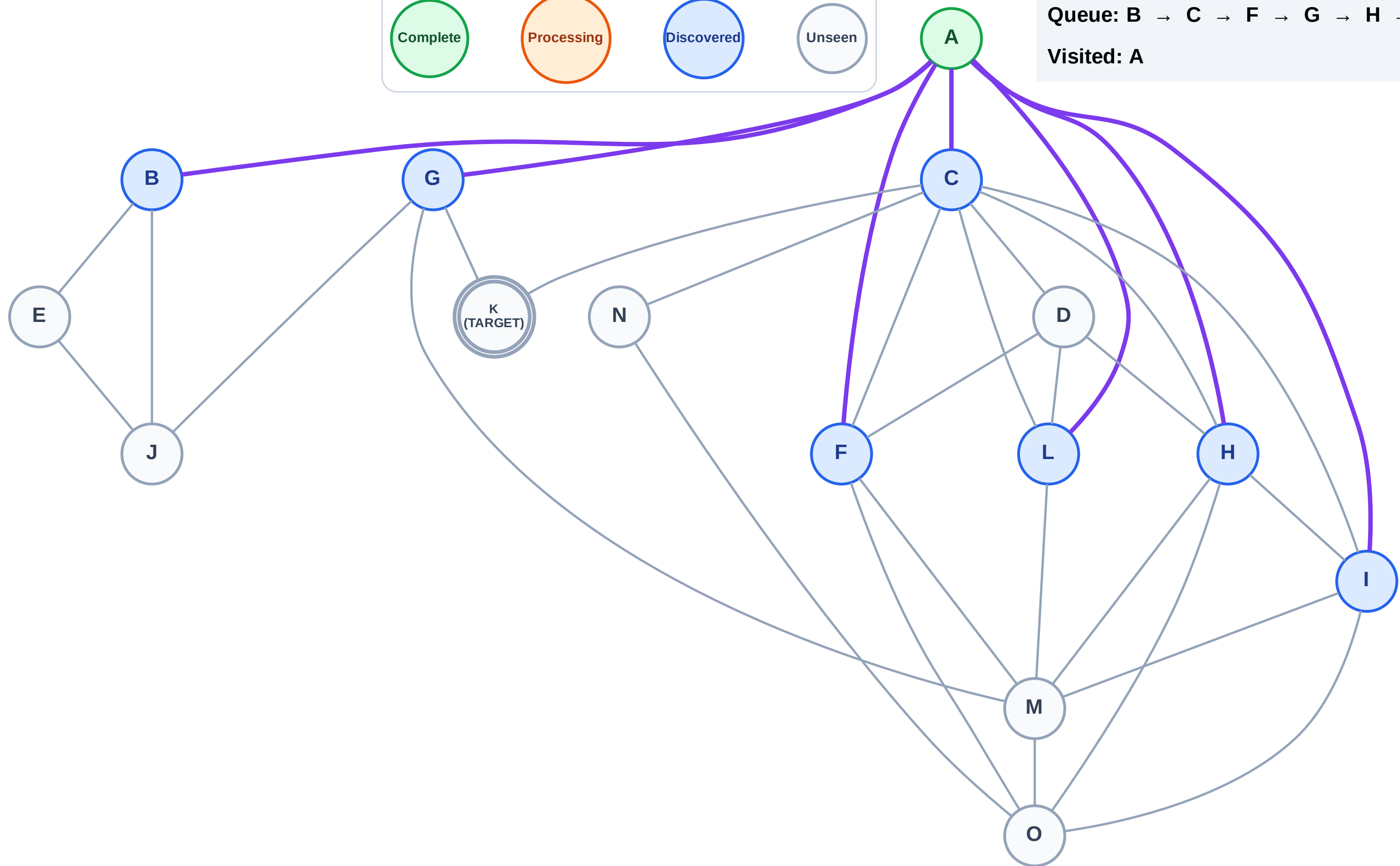
BFS Traversal

Step 3: Discover B, C, F, G, H, I, L from A

Legend



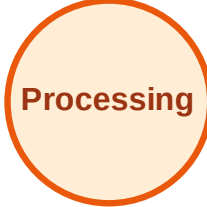
Queue: B → C → F → G → H → I → L
Visited: A



BFS Traversal

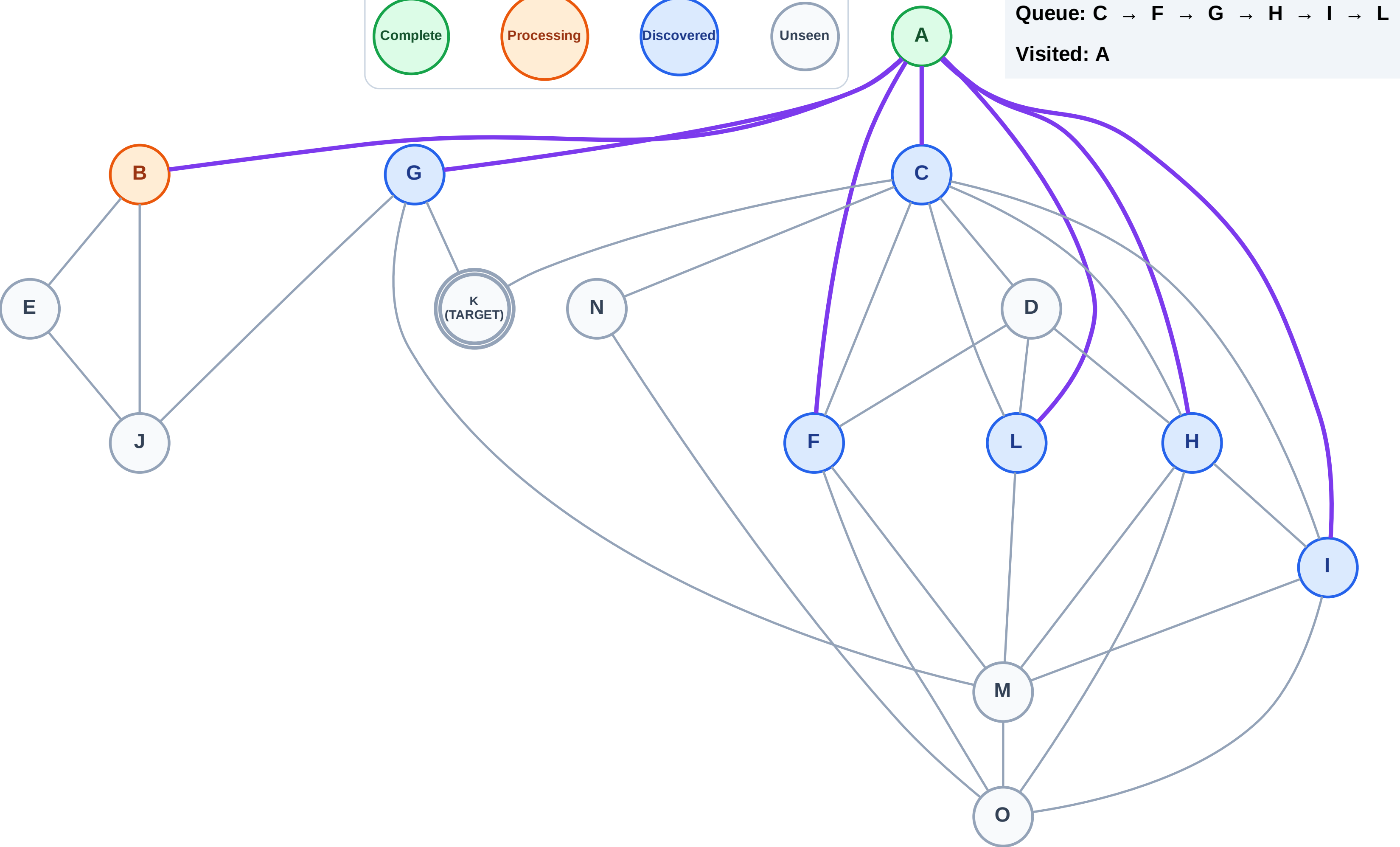
Step 4: Process node B

Legend



Queue: C → F → G → H → I → L

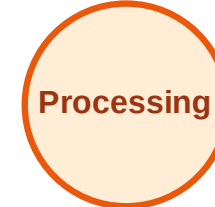
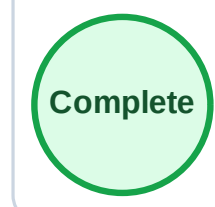
Visited: A



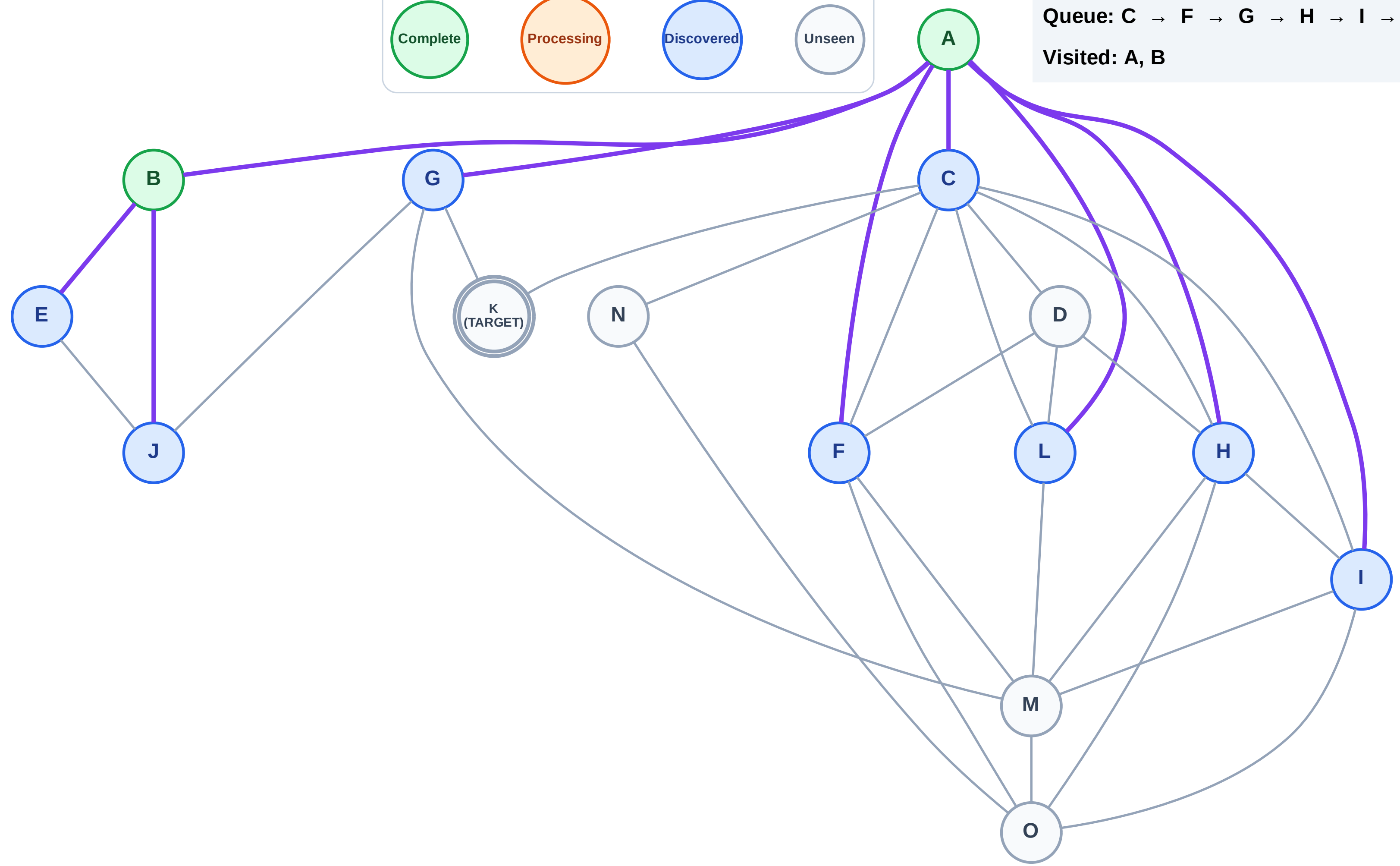
BFS Traversal

Step 5: Discover E, J from B

Legend



Queue: C → F → G → H → I → L → E → J
Visited: A, B



Step 6: Process node C

and

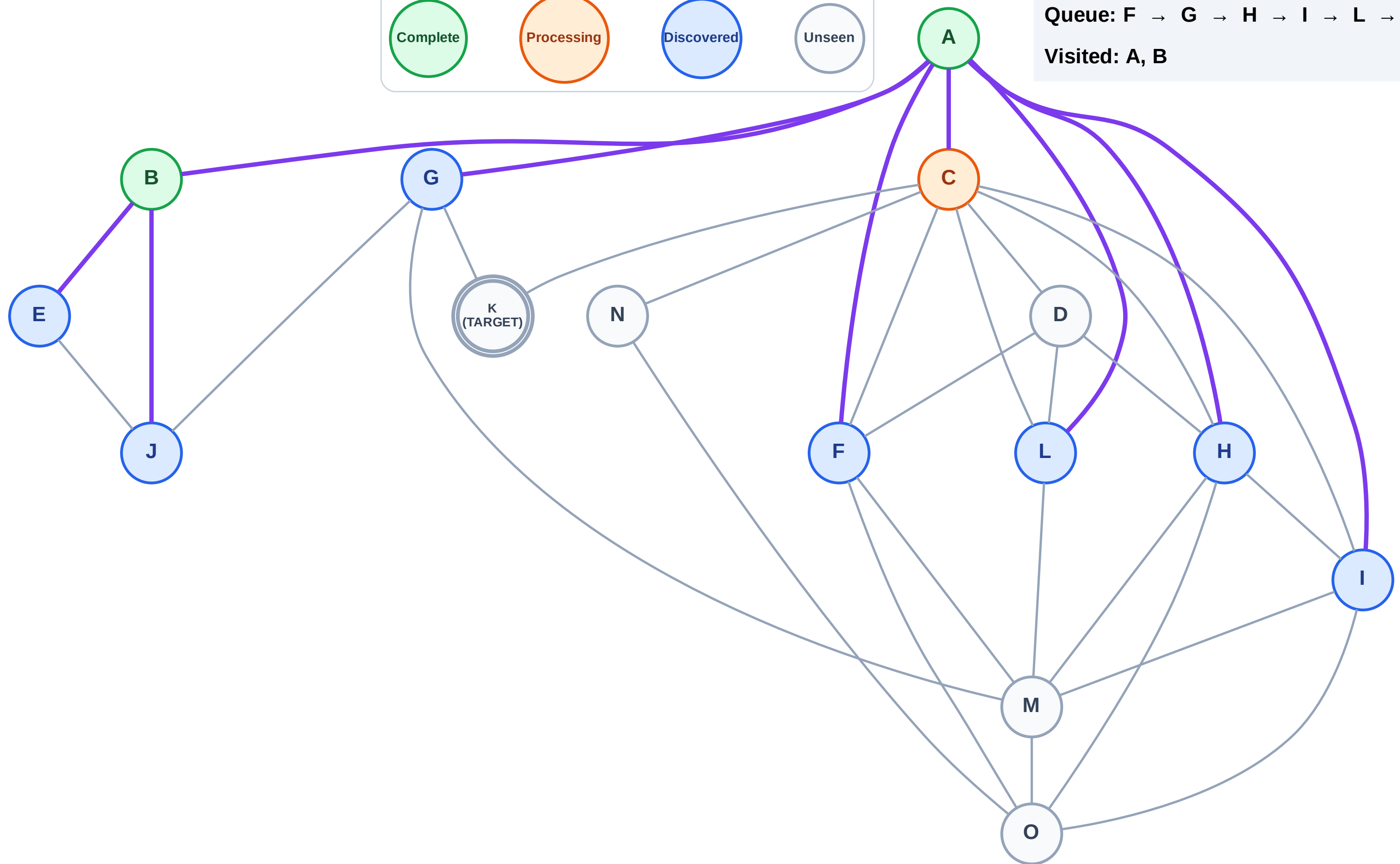
Legend



The legend consists of four colored circles arranged horizontally, each containing a label. From left to right: a green circle with the label 'Complete', an orange circle with the label 'Processing', a blue circle with the label 'Discovered', and a grey circle with the label 'Unseen'.

Unseen

Visited: A, B



BFS Traversal

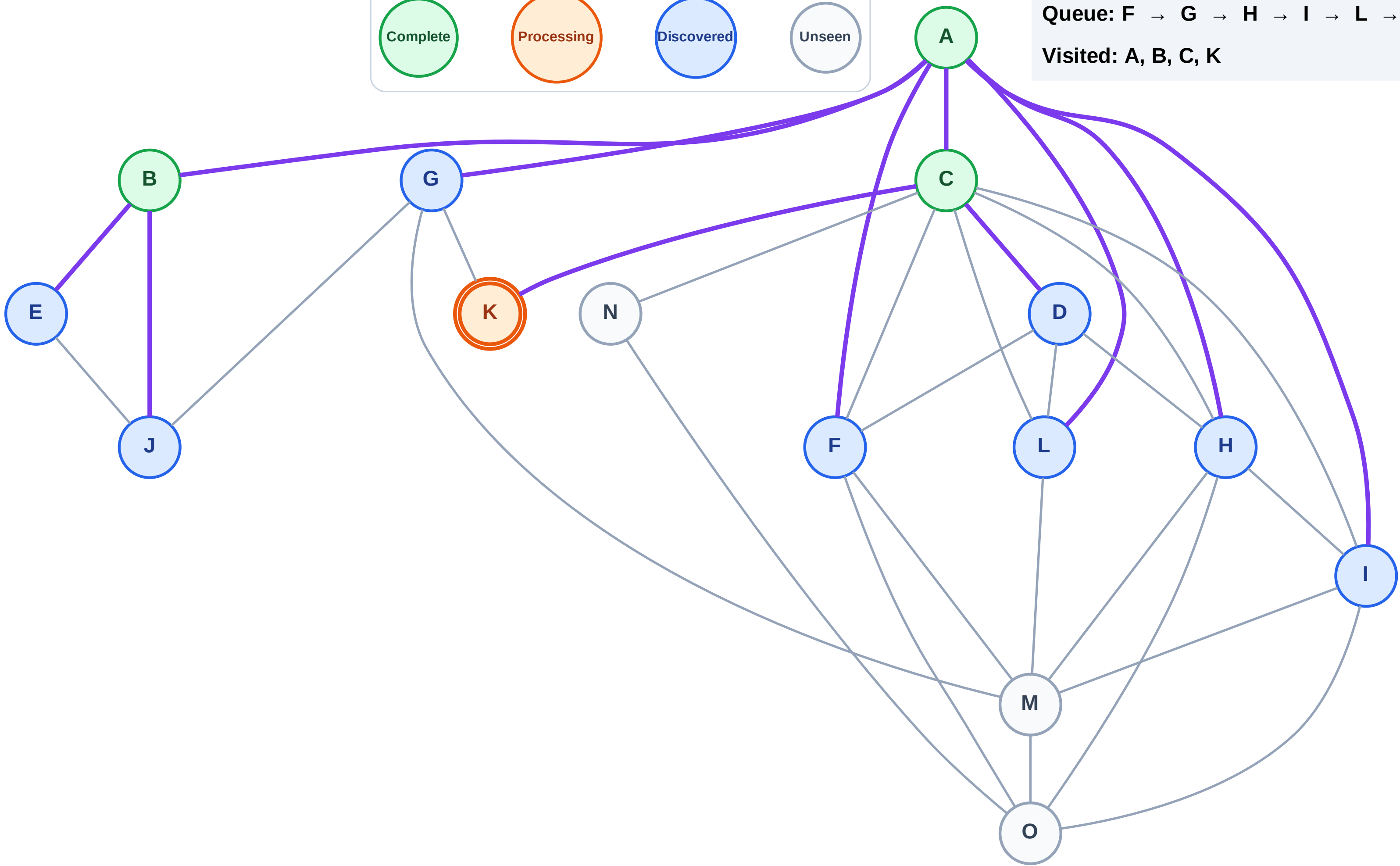
Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → C → K

Legend



Queue: F → G → H → I → L → E → J → D

Visited: A, B, C, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

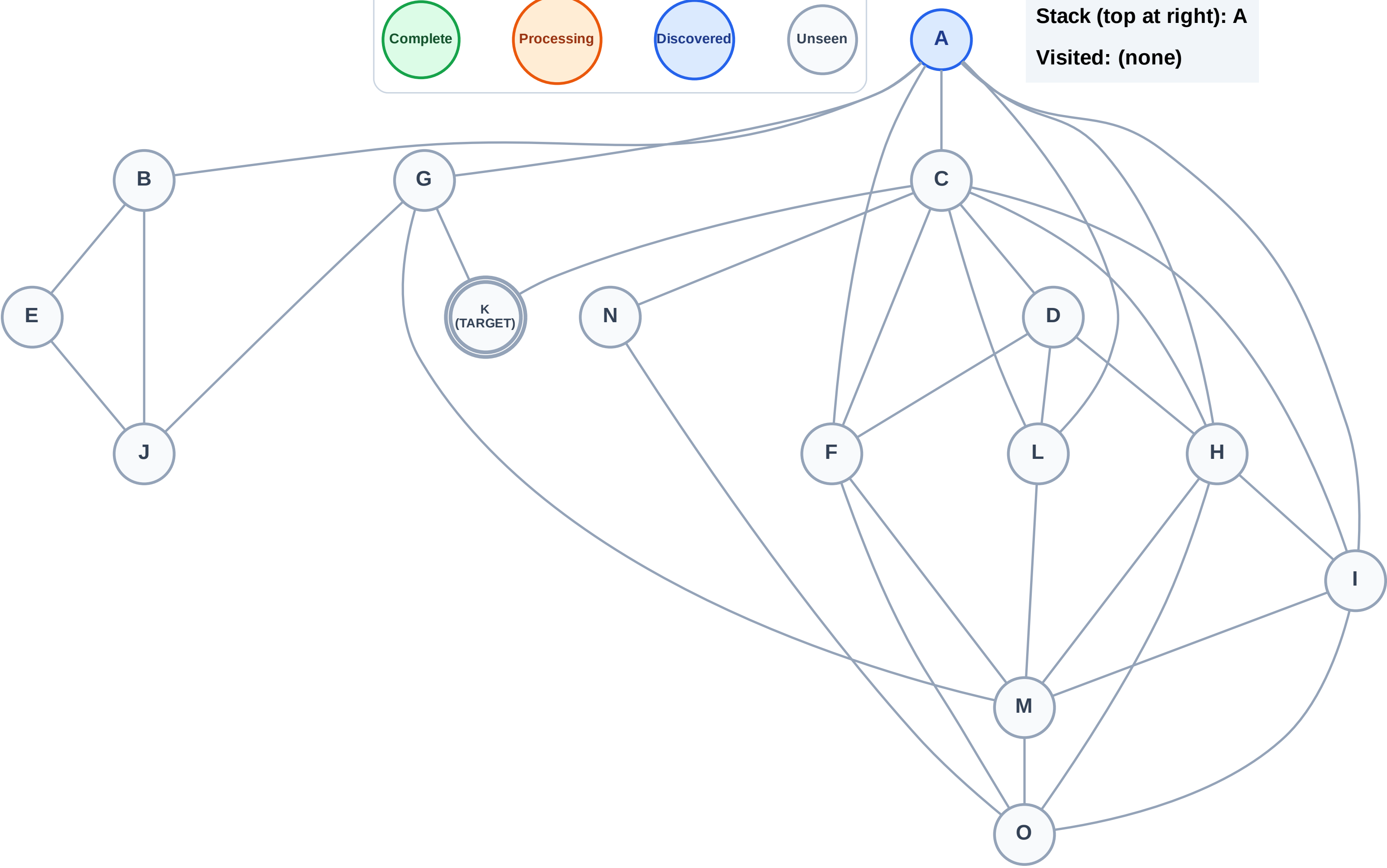
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

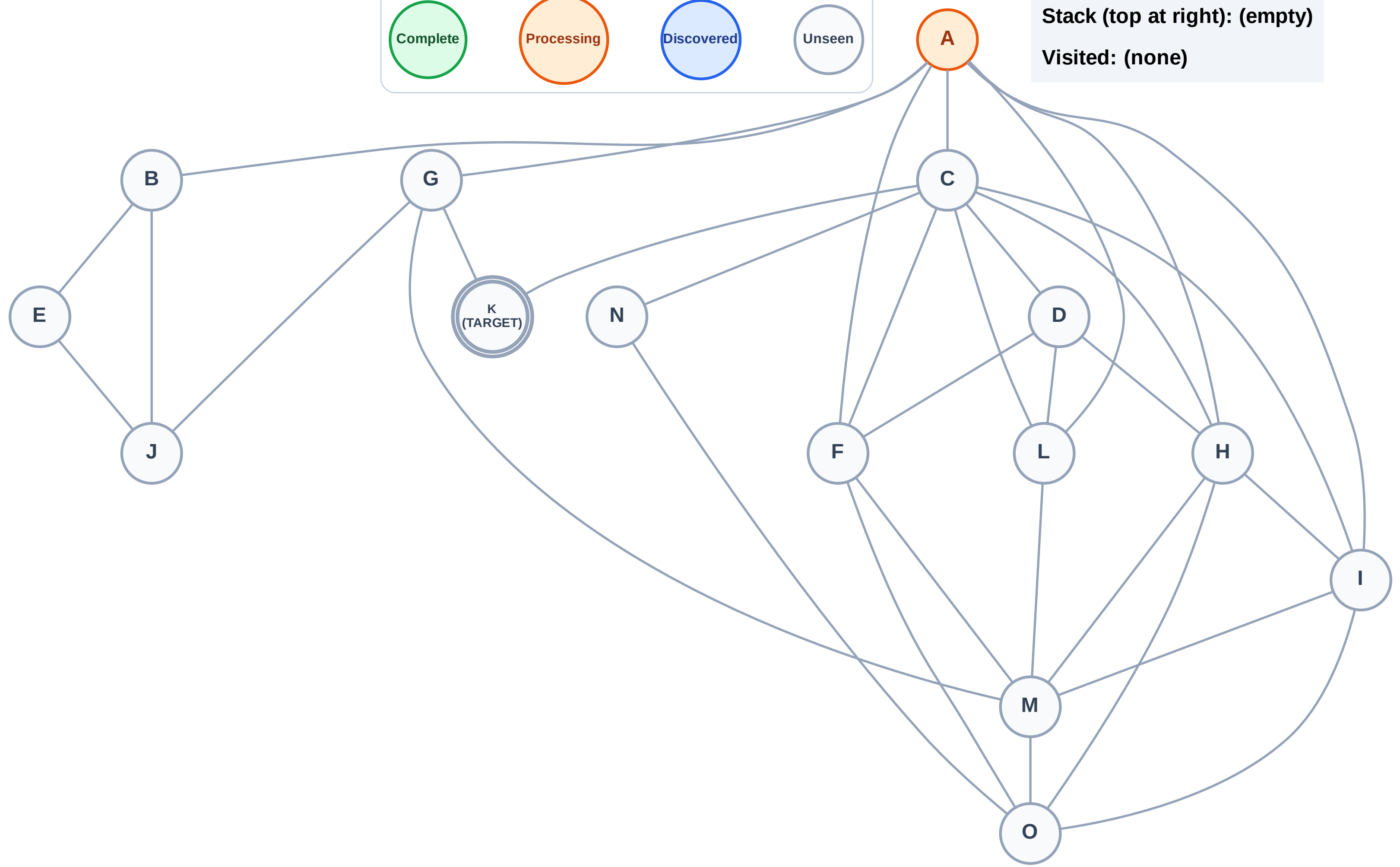
Step 2: Process node A

Legend



Stack (top at right): (empty)

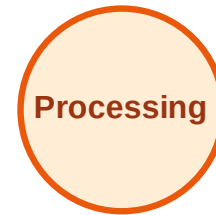
Visited: (none)



DFS Traversal

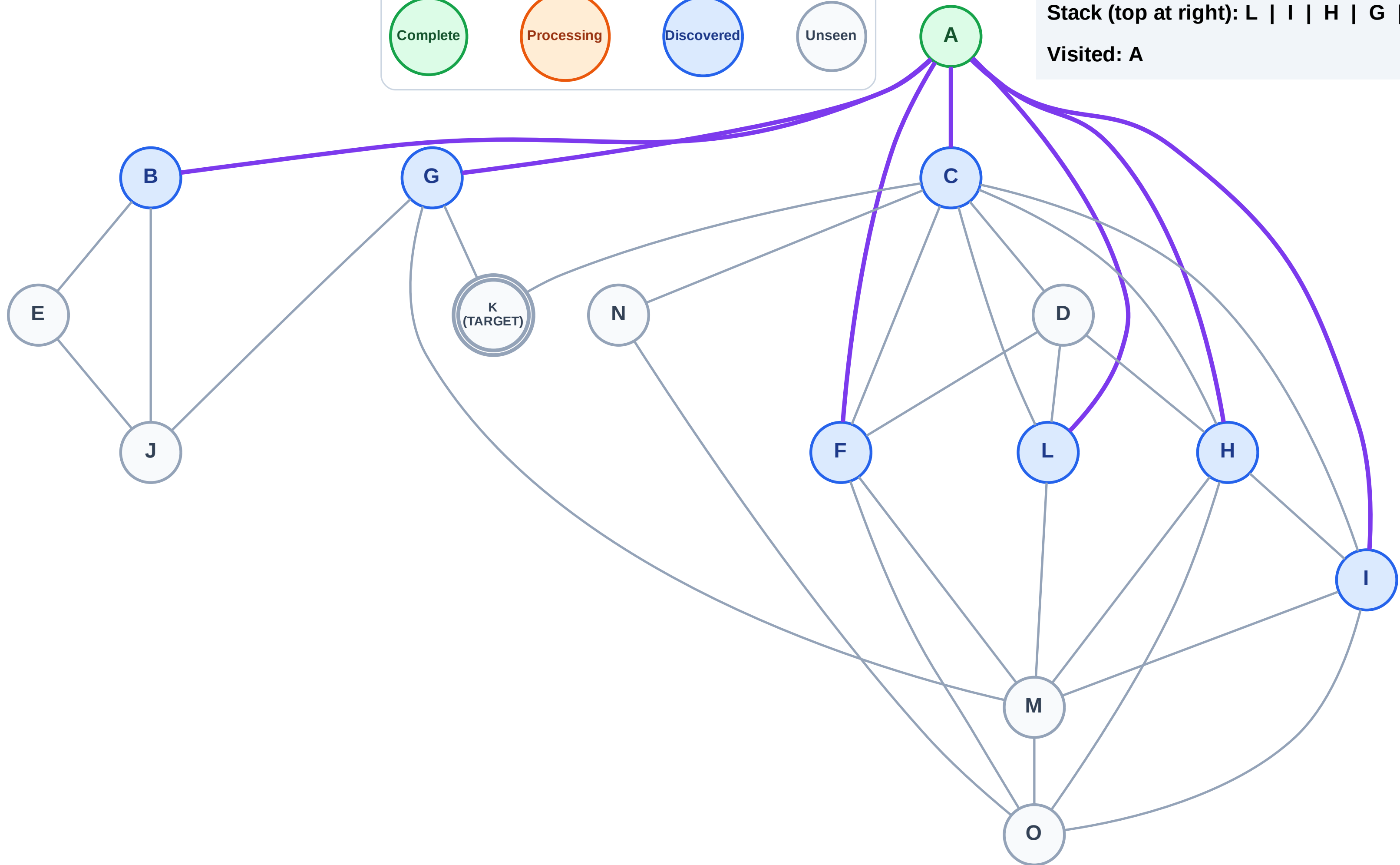
Step 3: Discover L, I, H, G, F, C, B from A

Legend



Stack (top at right): L | I | H | G | F | C | B

Visited: A



DFS Traversal

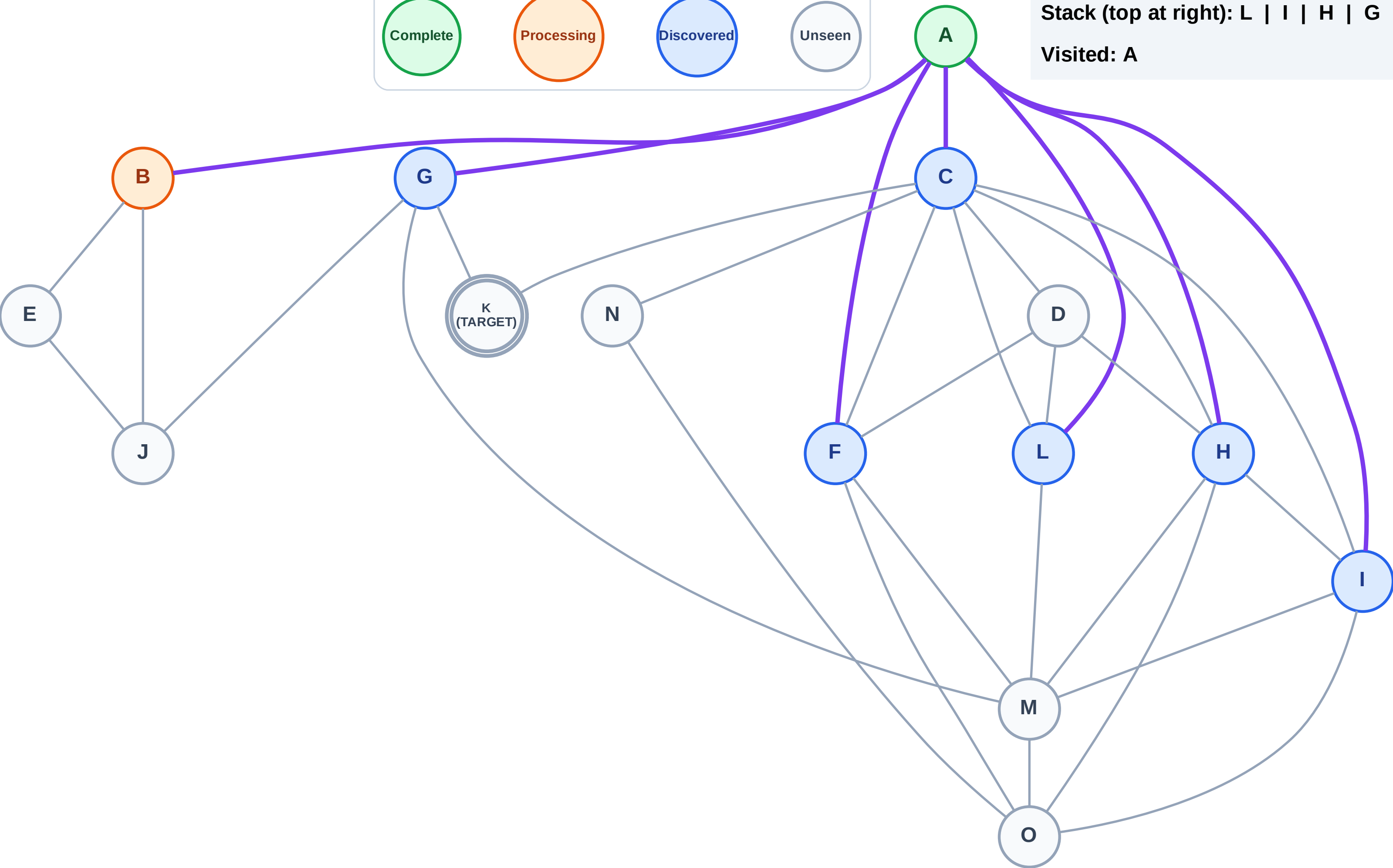
Step 4: Process node B

Legend



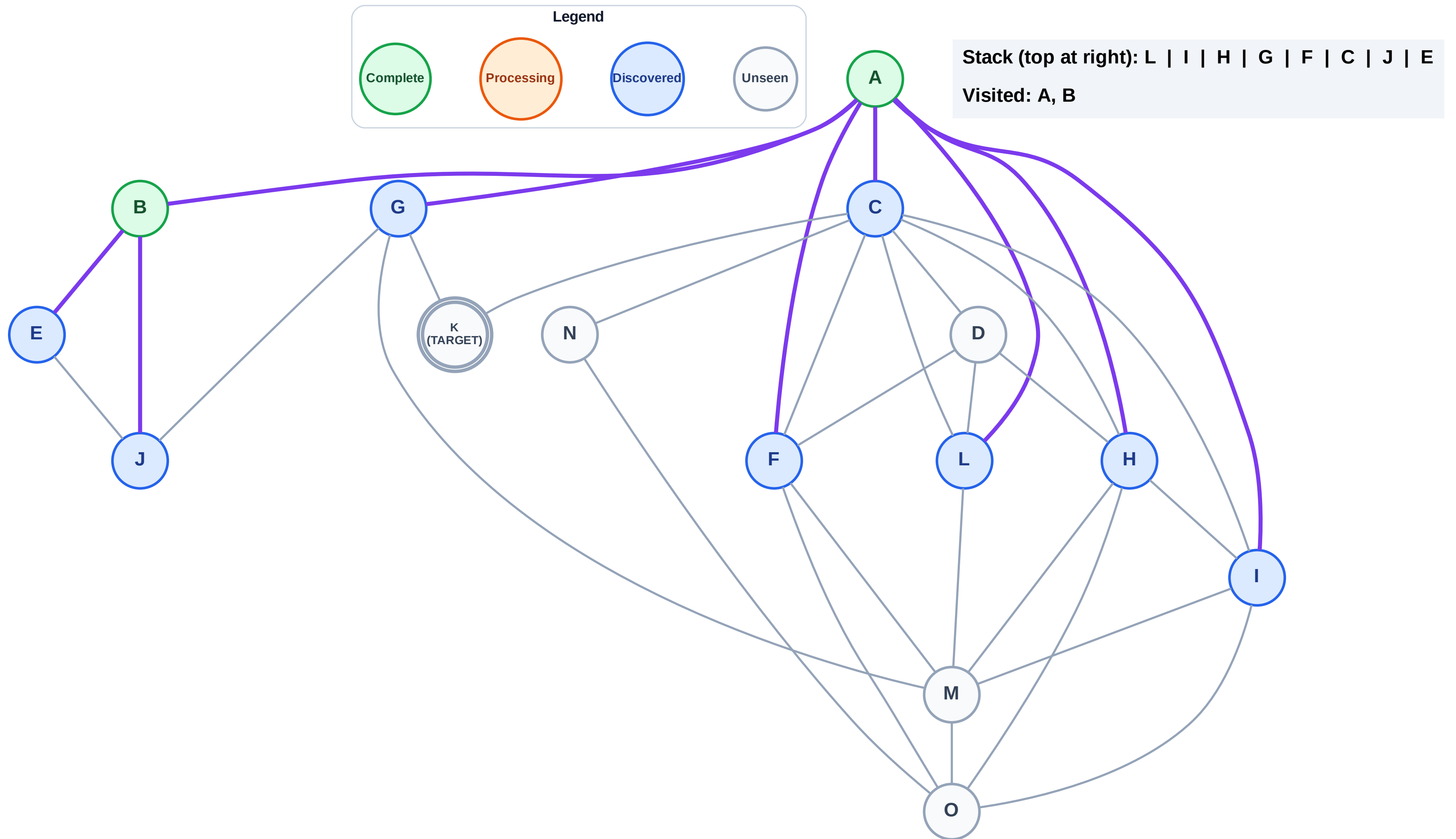
Stack (top at right): L | I | H | G | F | C

Visited: A



Step 5: Discover J, E from B

Step 5: Discover J, E from B



DFS Traversal

Step 6: Process node E

Legend



Complete



Processing



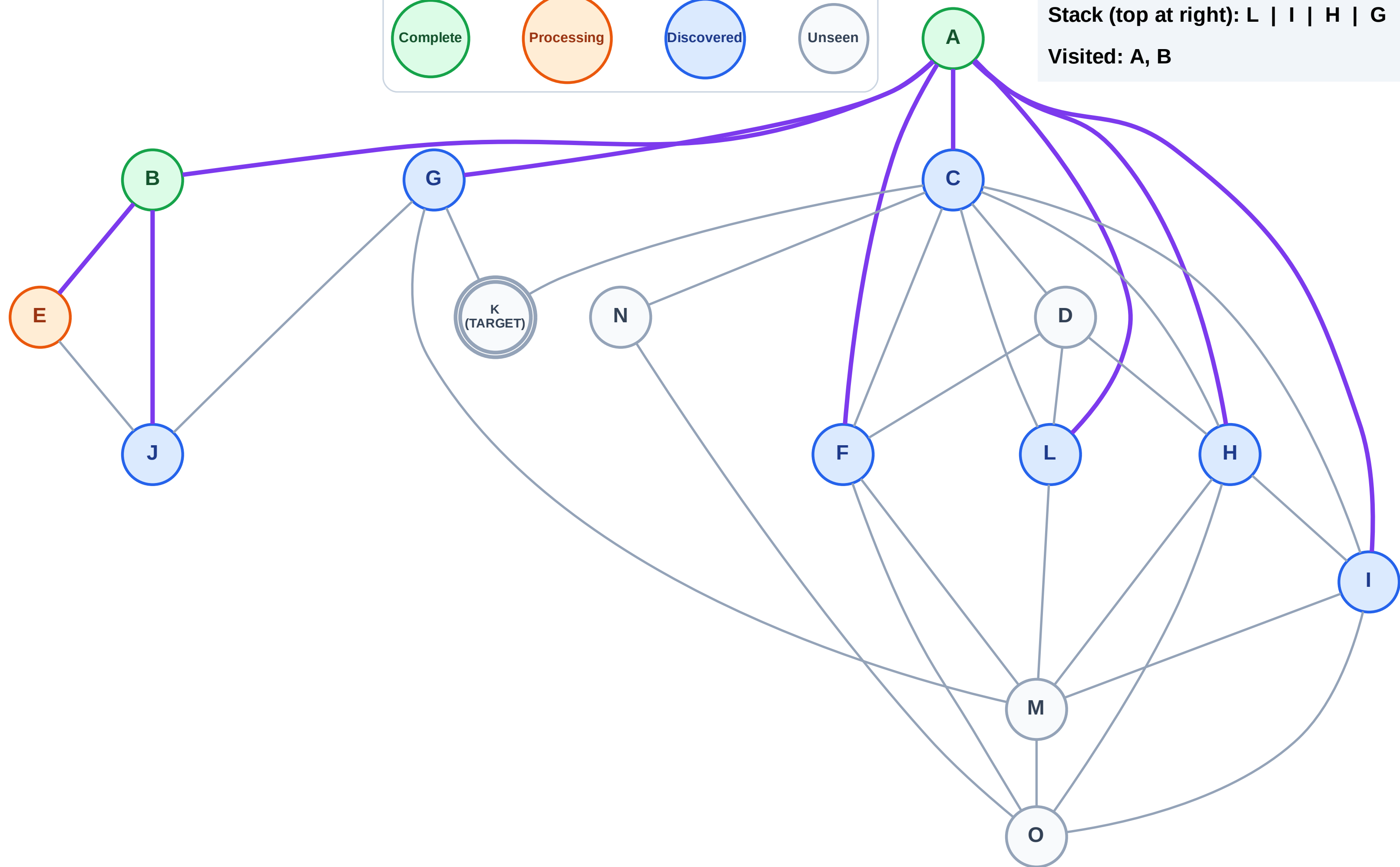
Discovered



Unseen

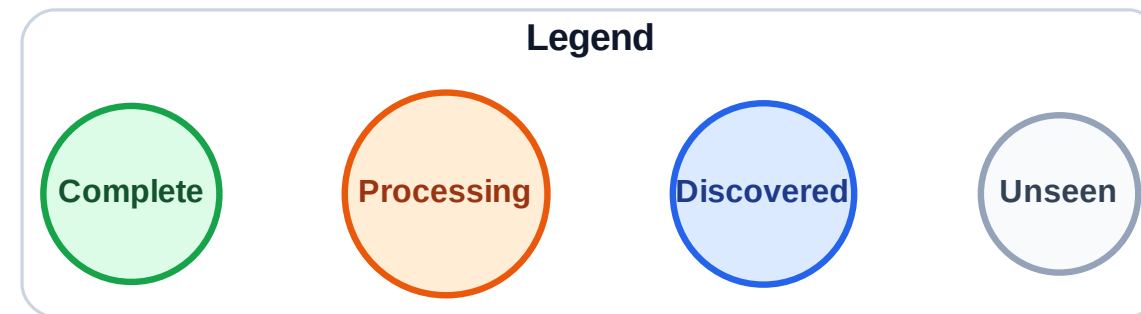
Stack (top at right): L | I | H | G | F | C | J

Visited: A, B



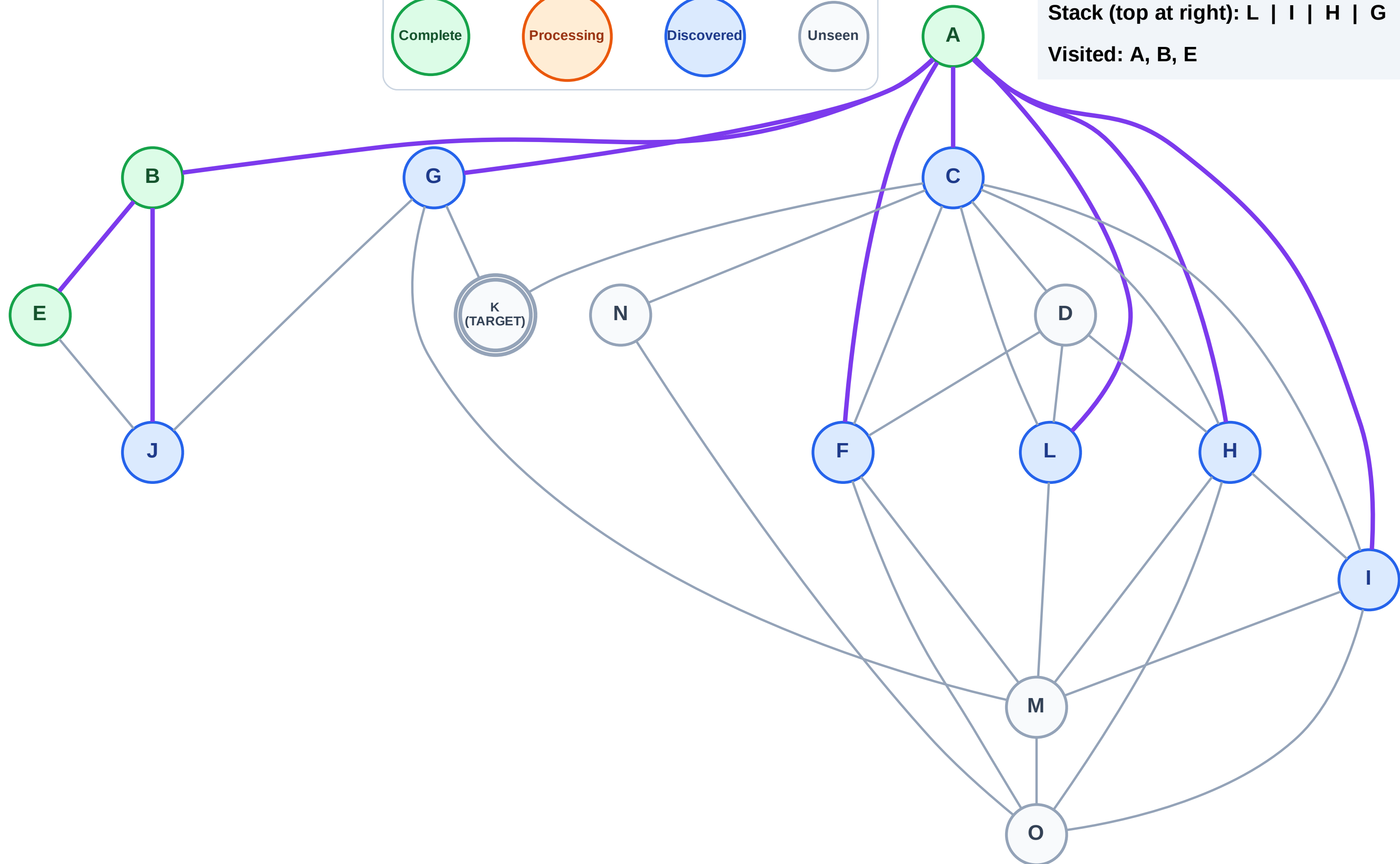
Step 7: No new neighbors from E

Step 7: No new neighbors from E



Stack (top at right): L | I | H | G | F | C | J

Visited: A, B, E



DFS Traversal

Step 8: Process node J

Legend

Complete

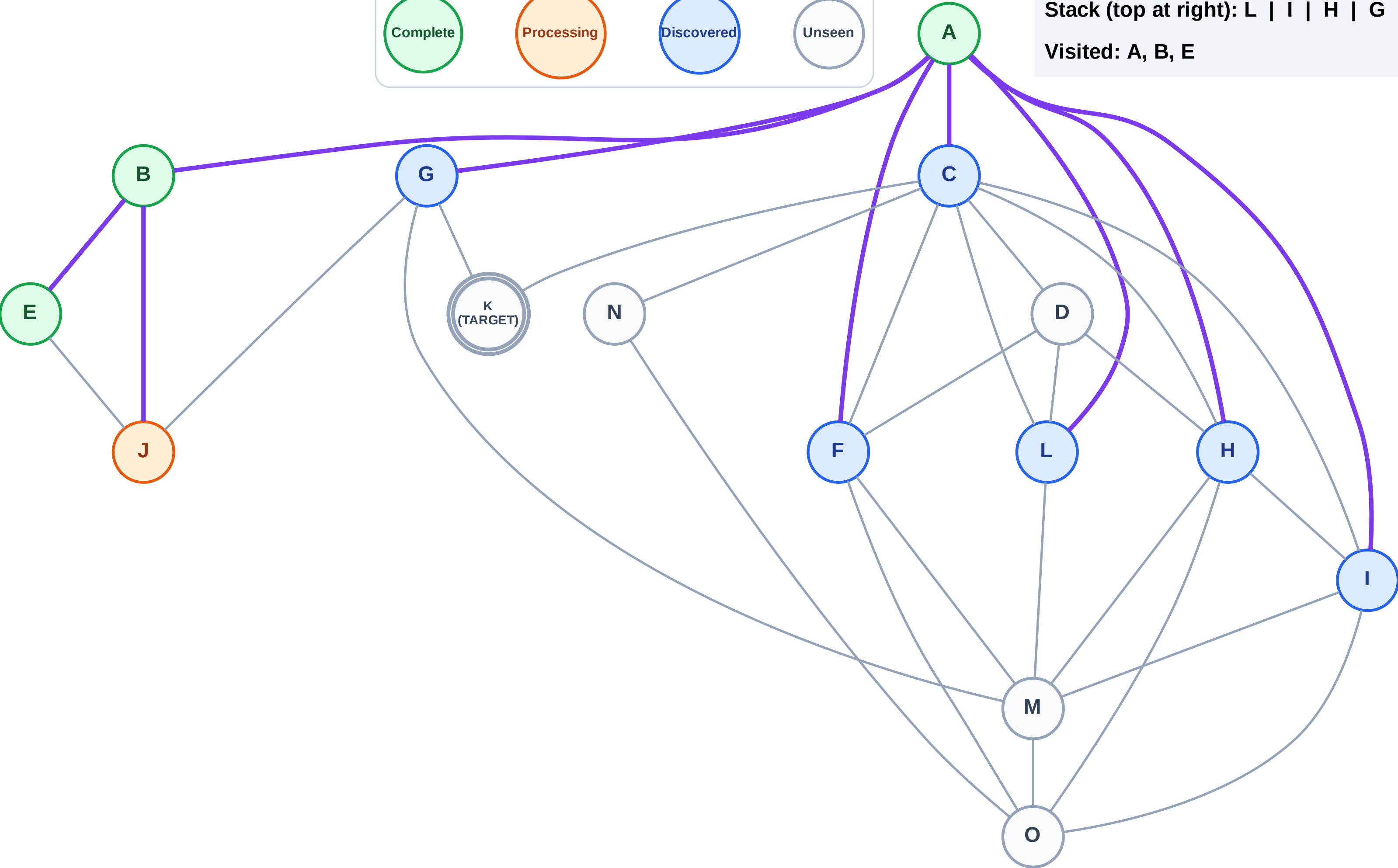
Processing

Discovered

Unseen

Stack (top at right): L | I | H | G | F | C

Visited: A, B, E



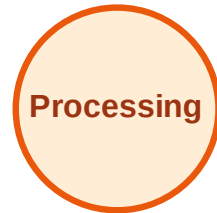
DFS Traversal

Step 9: No new neighbors from J

Legend



Complete



Processing

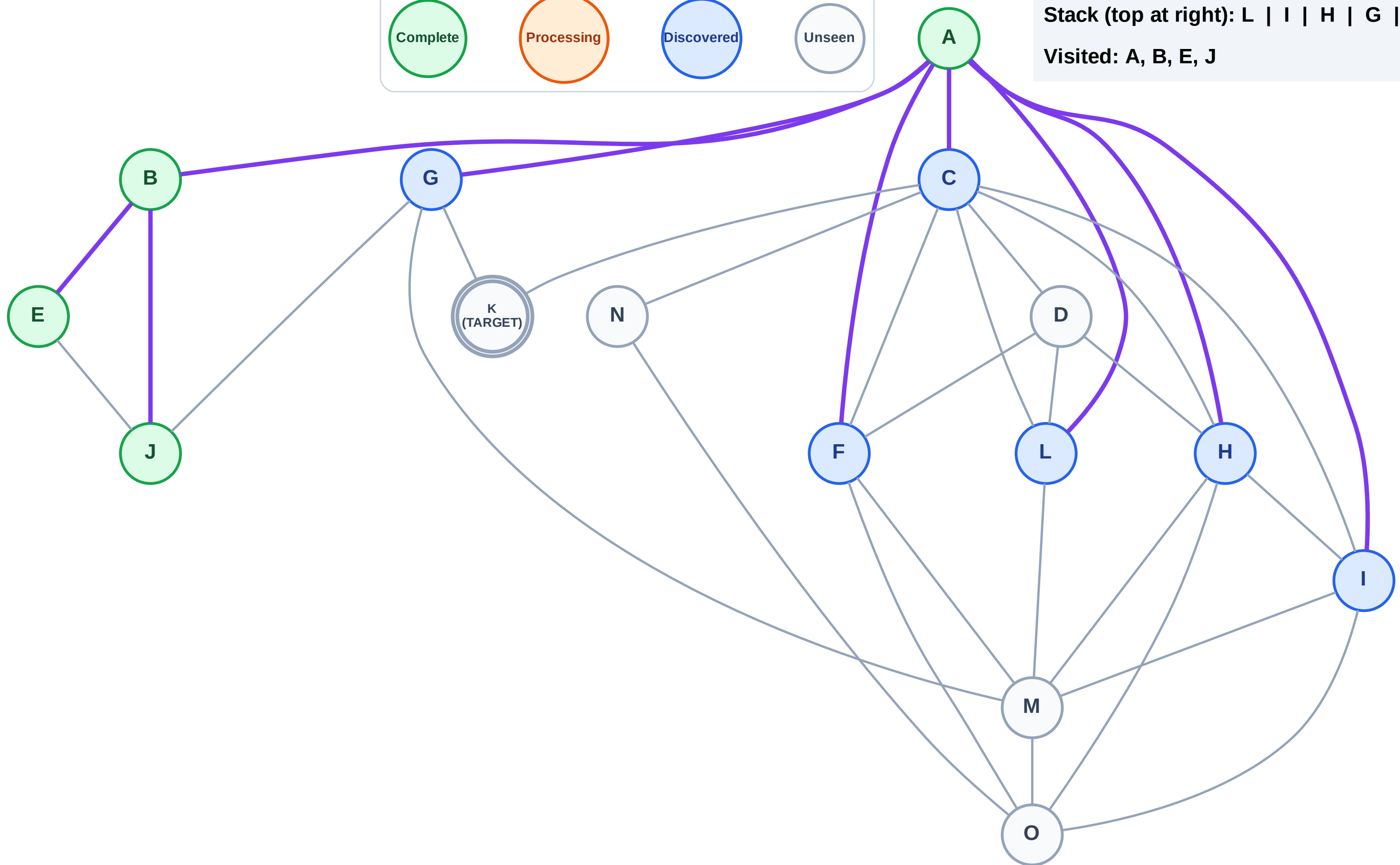


Discovered



Unseen

Stack (top at right): L | I | H | G | F | C
Visited: A, B, E, J



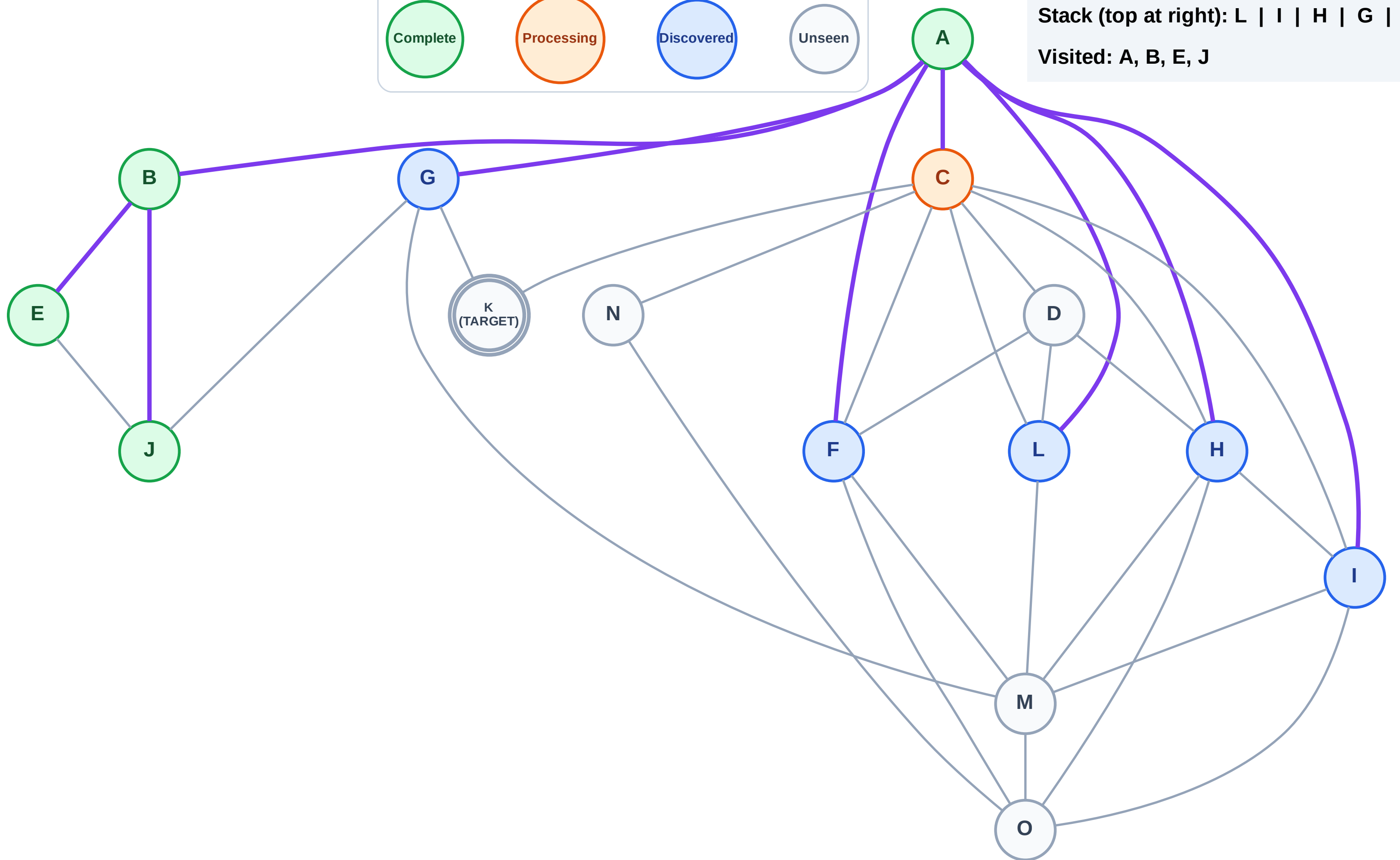
DFS Traversal

Step 10: Process node C

Legend



Stack (top at right): L | I | H | G | F
Visited: A, B, E, J



DFS Traversal

Step 11: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → C → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): L | I | H | G | F | N

Visited: A, B, C, E, J, K

